# Class 37: Electron Compounds; Phonons, Optoelectronic Materials

In this class we will look at two more instances where the theories developed so far in this course help us get additional insight into the properties of materials. First we will look at electron compounds, where a phase change is explained using the theories we have developed in this course, and then we will look at phonons, their properties and the role they play in the interaction between light and certain types of semiconductors.

**Electron Compounds:**
The success of the theories we have developed so far can be gauged from the number of experimentally observed phenomena they are able to explain. In particular, we have concentrated on theories that apply to solid metallic systems, which contain considerable amounts of free electrons. Therefore, the theories developed are found to be generally effective in explaining properties that are dependent on the nearly free electrons. Some of these properties are the more commonly observed properties such as electrical conductivity, thermal conductivity, optical properties and magnetic properties. However, there are also other phenomena in materials, that are not so commonly encountered by us in day to day life and it is of interest to see if these too can be explained using the theories developed so far.

In this class, we begin by looking at a specific experimental phenomenon that is observed in some of the binary phase diagrams in the Ag and Cu systems.

Both Ag, and Cu have the FCC crystal structure. The terminal solid solutions in these systems also have the FCC crystal structure. On adding the alloying element which forms a substitutional solid solution with the solvent, up to a certain composition, the structure remains FCC. After this the structure changes to BCC. The exact composition at which the structure changes from FCC to BCC, varies depending on the solvent atom.

These binary systems were examined to determine the reason for the phase change. A natural valency could be associated with each of the elements used as the solute. Based on the valency of the solvent and the valency of the solute, and the molar ratio of the solid solution, it is possible to assign an ‘e/a’ ratio, which refers to the average number of valence electrons per atom in the alloy. Hume Rothery discovered that regardless of the alloying element used, the phase change occurs at a fixed value of the e/a ratio. Since the different solute atoms could have different valencies, the same e/a ratio occurs at different compositions in each of the different systems studied. These compounds which occur as a result of the valence electron to atom ratio, are called ‘electron compounds’ or ‘Hume Rothery phases’. While the experimental observation was useful, it still does not by itself explain why the phase change must occur at those specific e/a ratios.

It is of interest to see if the theories we have developed are able to explain the formation of electron compounds.

Consider the plot of the density of available states as a function of energy, as shown in Figure 37.1 below. In the case of nearly free electrons in a solid at room temperature, the free electron parabola is distorted at the Brillouin zone boundaries. Further, due to the effect of the

temperature, the highest occupied energy levels are spread out in energy and do not occur at a single specific value as they do at the $E_f$ at zero Kelvin.

Figure 37.1 below shows the density of occupied states as a function of energy for the FCC structure as well as for the BCC structure. For both Ag, and Cu, this schematic can be assumed to apply.

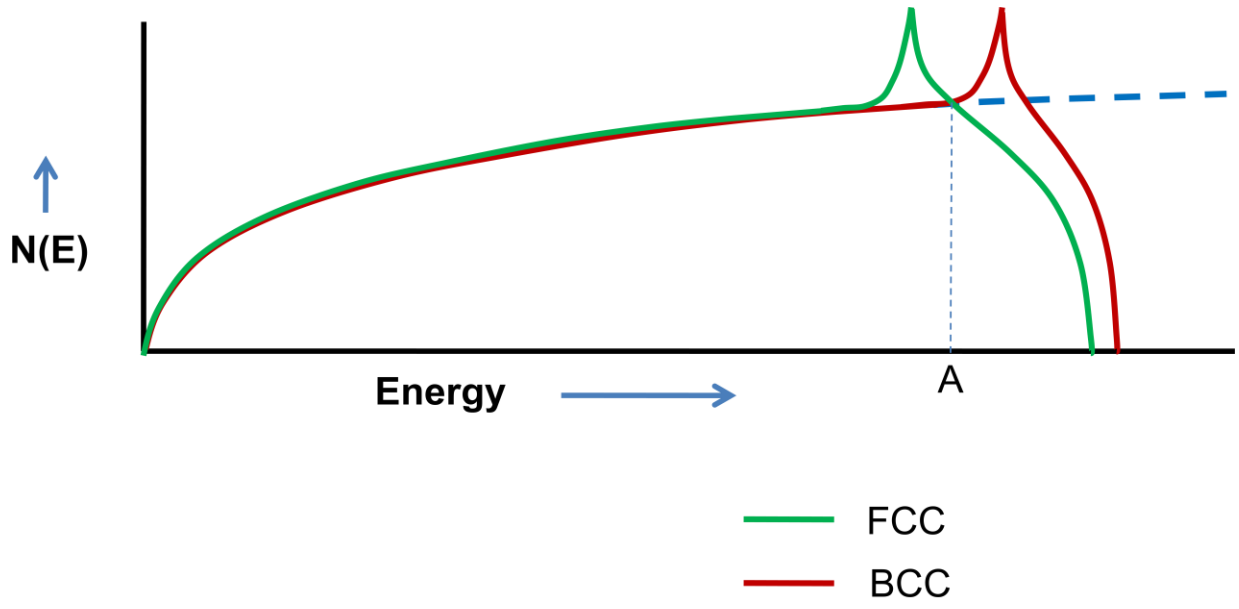



**Figure 37.1:** Schematic showing the density of occupied states as a function of energy for the FCC structure as well as for the BCC structure. This schematic can be assumed to apply for both Ag, as well as for Cu

Near the Brillouin zone boundaries, with increasing energy, the density of occupied states increases as a function of energy and then decreases sharply. From Figure 37.1 above, it can be seen that since the Brillouin zone boundary of the BCC structure occurs at a larger value of energy, the distortion of the free electron parabola occurs at a higher level of energy in the BCC structure, compared to the FCC structure. As can be seen from the figure above, at energy levels greater than an energy level identified on the energy axis by the point 'A', the $N(E)$ for the FCC structure is less than that of the BCC structure, and continues to decrease, while the $N(E)$ for the BCC structure continues to increase for some more values of energy. Therefore, if the number of electrons in the system is such that the $E_f$ is greater than that corresponding to the point 'A', the nearly free electrons in the system can be accommodated in lower energy states if the structure is BCC, than if it is FCC. Since there are a large number of nearly free electrons in the system, the difference in energy is significant enough to cause the structure to change, once a threshold level of electrons is exceeded. This concentration is expressed in the form of the nearly free electrons to atom ratio, and when this e/a ratio exceeds 1.4, the phase change occurs. Figure 37.2 below schematically shows the situation for e/a ratios of 1, 1.4, and 1.5. At an e/a ratio of 1.5, the electrons are held at lower energy levels in the BCC structure compared to the FCC structure. Hence the phase change occurs.

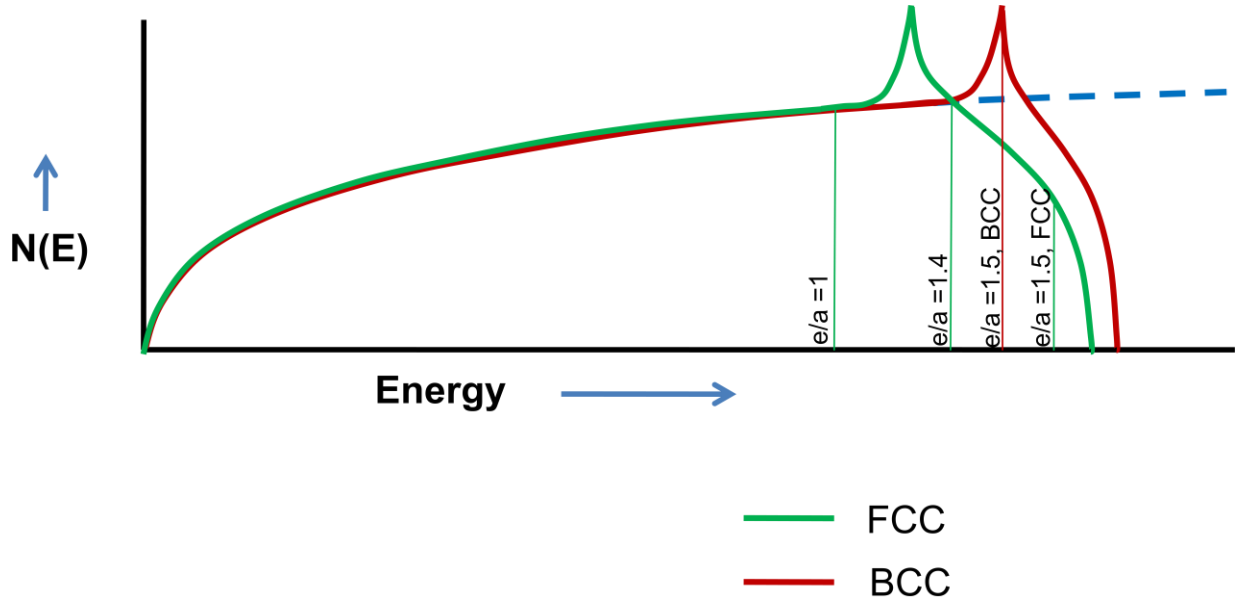



**Figure 37.2:** Schematic showing the effect of different e/a ratios on the highest energy level occupied in the FCC structure and in the BCC structure. At e/a ratios greater than 1.4, electrons are held at lower energy levels in the BCC structure in comparison to the FCC structure – an example of e/a=1.5 is shown for illustrative purposes.

**Phonons:**
In this course, we have so far focused largely on the electrons in the materials and ignored the atoms, or the ionic cores. Atoms can vibrate about their mean positions and can gain or release energy, or in other words, participate in the energy transaction process. Atoms vibrating about their mean positions result in waves of lattice vibrations, that can travel through the solid at $v_s$, the speed of sound.

The form of the wave that travels through the solid can be arbitrary, but it is possible to write it down as a sum of waves that are well defined and sinusoidal, as obtained using a fourier series. Therefore each wave can be identified with specific wavelength and frequency.

Waves in matter, or waves of lattice vibrations, are called phonons.

Taken in its entirety, the specific heat of a solid has a significant contribution from phonons, and this represents the atomic contribution to the specific heat – so far we have only looked at the nearly free electron contribution to specific heat.

One of the successes of quantum mechanics is that phonons too behave consistent with quantum mechanical rules.

For phonons, in just the manner we saw for photons,

$$E = h\nu$$

In other words, the energy of phonons is also quantized. One major difference between photons and phonons is that the former travel at the speed of light, while the latter travel at the speed of sound. Therefore, for phonons,

$$\nu = \frac{v_s}{\lambda}$$

where $v_s$, as we indicated earlier, is the speed of sound in that material.

Phonons can therefore be thought of as lattice wave equivalent of photons.

In a solid of length $L$ and inter-atomic spacing $d$, the maximum wavelength that can be supported is $2L$, in which case the neighboring atoms are displaced in the same direction from their mean positions, and the minimum wavelength that can be supported is $2d$. To support specific wavelengths in between these two limits, some adjacent atoms will move in opposite directions.

In case the solid has a two atom basis, and if the two atoms are ionically bonded, then due to the specific wavelengths where the atoms move in opposite directions, an electromagnetic wave is generated. In this case the phonons are able to interact with incident electromagnetic radiation of corresponding frequencies. The typical range of such frequencies places them in the infra red region of the electromagnetic spectrum.

Phonons that can interact with electromagnetic radiation, are referred to as ‘Optically active’ phonons. The rest of the phonons are referred to as ‘Acoustically active’ phonons.

As indicated in an earlier class, phonons become necessary to enable electron transitions in indirect band gap semiconductors such as Si, but are not required to enable transitions in direct band gap semiconductors such as GaAs. Therefore direct band gap semiconductors are preferred for opto-electronic materials which depend on electrical as well as optical properties for their functioning. The functioning of indirect band gap semiconductors is highlighted in Figure 37.3 below.

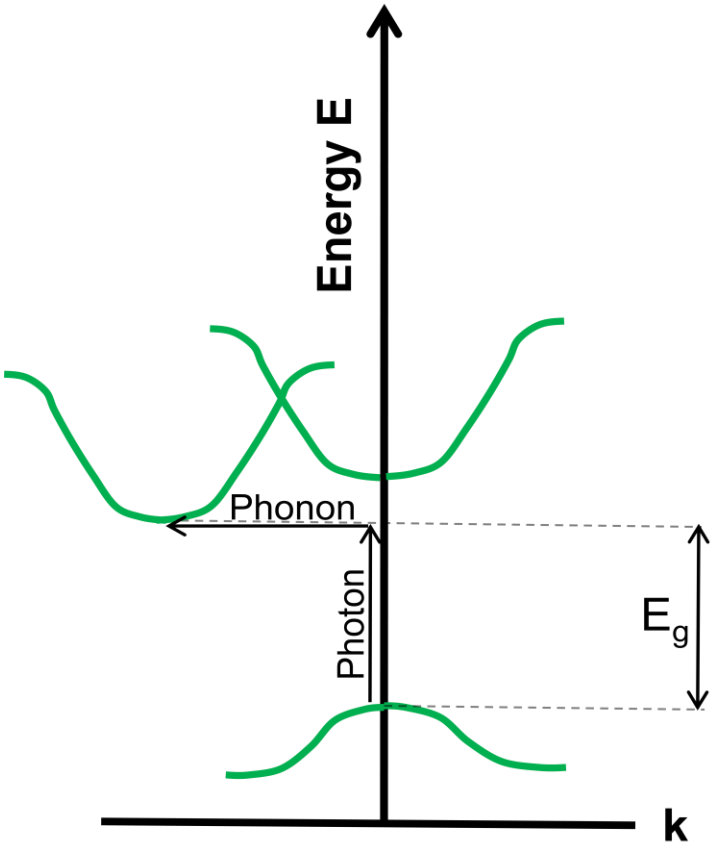



**Figure 37.3:** The photon followed by phonon process required for an indirect band gap semiconductor to absorb incident light.

In just the manner in which we looked at the statistical behavior of electrons, and found that they are Fermions, photons and phonons follow a statistical behavior that is credited to Satyendra Nath Bose and Albert Einstein, and is called the Bose-Einstein statistics. Photons and phonons are therefore referred to as Bosons.

In this class we have seen more material phenomena explained by the theories discussed so far. In the next class we will look at superconductivity as a phenomenon, which also involves another Boson. In the class after that we will derive the Bose-Einstein statistics.